Linguistic and Acoustic Features Depending on Different Situations - The Experiments Considering Speech Recognition Rate

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Abstract

This paper presents the characteristic differences of linguistic and acoustic features observed in different spoken dialogue situations and with different dialogue partners: human-human vs. human-machine interactions. We compare the linguistic and acoustic features of the user's speech to a spoken dialogue system and to a human operator in several goal setting and destination database searching tasks for a car navigation system. It has been pointed out that speech-based interaction has the potential to distract the driver's attention and degrade safety. On the other hand, it is not clear enough whether different dialogue situations and different dialogue partners cause any differences of linguistic or acoustic features on one's utterances in a speech interface system. Additionally, research about influence of speech recognition rate is not enough either. We collected a set of spoken dialogues by 12 subject speakers for each experiment under several dialogue situations. For a car driving situation, we prepared a virtual driving simulation system. We also prepared two patterns where we have two dialogue partners with different speech recognition rate (100% and about 80%). We analyzed the characteristic differences of user utterances caused by different dialogue situations and with different dialogue partners in two above mentioned patterns.

1. Introduction

In recent years, some spoken dialogue systems have been developed and used as practical applications as car-navigation systems or robots. However, their usability contains some problems such as inability to recognize spontaneous speech, difficulties in dealing with spoken corrections, and understanding the changes of dialogue turns. These take place because speech recognition and interpretation systems are easily affected by the user's speech style, so it is important to analyze user's behavior in different circumstances in order to improve the system performance.

In the study of user's behavior, Amalberti et al.[1] compared behavior of two groups (those who talk with a computer and those who talk with an operator) using WOZ-system. They reported that with a computer, the users used simpler expressions, used fewer words per dialogue, and regarded the system as a tool. On the contrary, the users talked to the operator in a cooperative manner. In the study of user's behavior with the circumstance considered, T.Itoh et al.[2] prepared two dialogue partners (a spoken dialogue system and a human operator) and concurrent task (a car-driving task) as dialogue situations. They combined the partners and situations, and compared linguistic and acoustic features of user's utterances under each condition. As a result, they reported that some acoustic and linguistic features were affected by whether the dialogue partner is a human or a machine, while some acoustic features alone were affected by a concurrent task. K.Itou et al.[3] also investigated dialogue characteristics in different communication modes. As a result, they also reported that some acoustic and linguistic features were affected by whether the dialogue partner is a human or machine. However, in their study, the features in the various situations are not enough investigated, or the circumstances where they researched essentially include several different conditions such as speech recognition rate or response capability, and it is not clear what factor actually affects user's utterances.

This study analyzes the influence of different dialogue situations with the linguistic and acoustic features on ground of speech recognition rate and response capability. The different dialogue situations include the change of the dialogue partners - a human, a machine and a human who is allowed to do only what a machine can do. We experimented in the three situations with car simulation task and without it. While acoustic change in spoken corrections have been reported[5, 6, 7], we also analyzed the influence of the above dialogue situations on their characteristics.

2. Change of Linguistic and Acoustic Features

2.1. Dialogue Data Collection

For investigating linguistic and acoustic properties and variability in different situational contexts in a task-oriented spoken dialogue, we collected user's utterances through a series of experiments. The experiments were prepared in two patterns; one is that the speech recognition rate of dialogue partners is 100% (*EXP1*) and the other is about 80% (*EXP2*). This is because speech recognition rate of human is nearly 100% while in the case of current spoken dialogue systems it ranges from 70% to 85%. For realization of constant speech recognition rate, a human listen to users' utterance in all the situations and make a wrong response on purpose based on the steady probability. The dialogue task simulates voice control of a car navigation system, where one should perform goal setting by entering several goal names along a driving route. 12 kinds of driving route setting scenario are prepared. Each scenario includes three goals and one of them such as a hotel, a coffee shop and so on, must be looked up in the destination database. The subjects should memorize all the goals before each task and convey those names to the dialogue partner during the task.

In this experiment, two dialogue situations and three di-

alogue partners were prepared as shown in the Table 1. Two dimensions of situational settings are considered: existence/absence of a concurrent task and changes of the dialogue partners. In car-driving task, subjects have to drive at 100km/h along an oval-course. In order for subjects to focus on driving a car, before starting the experiment subjects are told to start over again if they fail driving. As for the dialogue partners, we prepared three patterns. One is a human operator, second is a spoken dialogue system and the last is a human operator who behaves just like a spoken dialogue system. Hereinafter, it is called pseudo-system.

Table 1: *Dialogue situations*

Partner		Operator		Machine
Voice		Natural	Synthetic	
Driving task(DT)	No	O	P	W
	Yes	O+DT	P+DT	W+DT

WOZ-system is used as a substitute for a spoken dialogue system. An operator for a role of WOZ listens to user's utterance and chooses an adequate response prepared previously. The response is conveyed to the user in synthetic voice. The users were explained that this system could accept any type of utterance and they could say whatever was associated with goal setting. By using WOZ-system, the speech recognition performance becomes variable.

The human operator speaks spontaneously but the utterance style as a response to user's utterance is restricted. In other words, the expressions which the operator is allowed to use are not the same as the system is allowed but the intention which is able to be conveyed to the user is the same. First of all in the task, the human operator introduces oneself in order to explain his status, which is normal in speaking to someone for the first time such as a call-center agent. It is expected that more natural utterances are collected through this process.

The Pseudo-system was operated by the same person as the human operator and by using the same application as WOZ-system. Thus, the only different point between Pseudo-system and WOZ-system is that the users know if the dialogue partner is a human or a machine.

The subjects were 12 male students and at the beginning of both experiments, they were made to practice driving in order to decrease the influence of their developing. And to keep results less affected, the order of dialogue situations and scenarios was decided at random.

2.2. Linguistic and Acoustic Features between Two Experiments

Table 2: *the Number of Utterances*

Dialogue Situation	O	O+DT	P	P+DT	W	W+DT
EXP1	236	237	227	227	227	227
EXP2	245	247	232	243	249	235

We compared the dialogue data of *EXP1* and *EXP2* except utterances of correction, reentry and negation. The reason is that because in *EXP1*, no recognition error exists and utterance as correction, reentry and negation - are not included, we cannot make a simple comparison. Table 2 shows the number of user's utterances which we collected in each situation. Table 3 shows the features of both experiments. All the acoustic statistics are

Table 3: *Linguistic and Acoustic features of between EXP1 and EXP2*

Experiment	*EXP1*	*EXP2*
Tasks	144	144
Utterances	1383	1451
Split utterance	187	265
Words/Utt	4.60	5.49
Filled pauses	446	341
Keywords/Utt	2.02	1.92
New-keywords/Utt	1.97	1.89
Omitted verbs	489	409
RMS mean	779	1018
RMS max	3151	4161
Pitch mean	75.4	76.5
Pitch min	58.7	57.6
Pitch max	269.4	263.2
Pitch S.D.	16.9	19.9

based on the following raw acoustic features: the speaking rate, the energy value as the RMS value and the fundamental frequency (RMS, Pitch). The speaking rate is measured in term of the number of morae per second. The "Duration(1)" and "Speaking rate(1)" features measure the utterance duration and speaking rate including the pause duration, while the "Duration(2)" and "Speaking rate(2)" feature measures exclude the pause duration. The pause means the duration, which standard is based on the RMS. According to the result of the T and F statistical tests, the following characteristics were found to be significant for the *EXP2* in comparison with the *EXP1*: decrease of the keywords and new-keywords per utterance ($p < 0.05$); increase of the split utterance ($p < 0.05$); decrease of the utterance except the verb ($p < 0.01$); increase of the words per utterance ($p < 0.01$); increase of the use of filled pauses ($p < 0.05$); increase in the mean and maximum of the RMS ($p < 0.01$); increase in the standard deviation of the pitch ($p < 0.01$);

To sum up the results, the users tend to talk with less information to those who sometimes hear wrong. Further, they do not omit the verbs and speak politely. As for acoustic features, the RMS increases and the intonation is emphasized. The reason for such results is that users change their utterances to make it easier for the dialogue partner to understand them.

2.3. Linguistic Features Caused by the Different Dialogue Partners

Table 4 and Table 5 show the statistics of linguistic features, which were separately calculated for each of six different dialogue situations. At the beginning, we compared the features between dialogue situations or dialogue partners in each experiment. According to the results of the T and F statistical tests, the following characteristics were found: increase of filled pauses in the case of a human operator than the others in both experiments; increase of verbs in the case of a human operator than the others in *EXP2* ($p < 0.05$); increase of words per utterance in the case of human operator than the others in *EXP2* without a concurrent task ($p < 0.1$);

The filled pauses which are used to pause to consider what to say are more used in the human partner. It may be that the result for more use of filled pauses in the case of the human partner is to insist that it is the user's turn or to achieve the time required to consider what to say next. However, in the case of the Pseudo-system, where the different point from the human partner is whether the voice is natural or synthetic, one tend not

Table 4: *Linguistic Features of EXP1*

Dialogue Situation	O	O+DT	P	P+DT	W	W+DT
Tasks	24	24	24	24	24	24
Utterances	236	237	227	227	227	229
Utterances/Task	9.83	9.88	9.46	9.46	9.46	9.54
Words/Utt	4.98	4.56	4.58	4.53	4.58	4.31
Filled pauses/Task	9.25	9.17	3.83	4.83	5	4.91
Keywords/Utt	1.99	1.97	2.07	1.98	2.07	1.99
New-keywords/Utt	1.97	1.91	2.03	1.96	1.96	1.95
Omitted verbs	71	70	86	85	88	89

Table 5: *Linguistic Features of EXP2*

Dialogue Situation	O	O+DT	P	P+DT	W	W+DT
Tasks	24	24	24	24	24	24
Utterances	245	247	232	243	249	235
Utterances/Task	10.20	10.29	9.67	10.13	10.38	9.79
Words/Utt	6.32	6.03	5.27	5.29	4.84	5.24
Filled pauses/Task	3.83	3.54	1.67	2.38	1.17	1.63
Keywords/Utt	1.96	1.93	1.96	1.84	1.85	1.98
New-keywords/Utt	1.95	1.89	1.94	1.82	1.81	1.93
Omitted verbs	38	41	79	77	93	81

to use the filled pauses even though knowing the partner is a real human being. Since synthetic voice gives an impression on users of speaking to a machine, rhythm elements of dialogue such as interrupt, overwrap and timing of beginning utterance (which are required generally in human-human dialogue) are more flexible. Thus, we think that there is little necessity of insisting users' turn so they less likely use filled pauses.

In the case of a human operator in *EXP2*, the words tend to be more used and the verbs tend to be used more frequently than the other dialogue partners. This fact suggests that if it is hard for users to convey their utterance to the partner, they tend to speak more politely when his voice is natural. We think it is because natural voice conveys the feeling, the speaker feels like talking in a polite way.

Next, we compared the features between the experiments in each condition and according to the results of T and F statistical tests, following characteristics were found for the *EXP2* in comparison to the *EXP1*: decrease of the keywords number in the case of Pseudo/Existence ($p < 0.05$) and WOZ/absence ($p < 0.1$); increase of the use of verbs in the case of Operator ($p < 0.1$); increase of the words in the case of Operator ($p < 0.05$);

From these results, it is found that the speech recognition rate of dialogue partner affects one's utterance. If the voice of partner is synthetic, users have a tendency to talk in fewer keywords, and if this voice is natural, they have a slightly strong tendency to talk more politely. It may be that the synthetic voice make users feel like talking to a machine so they try to speak in the way which they think a machine could recognize easily - use less keywords. And, in the case of human partner, because prosodic information that includes feeling is carried through the voice, users become more cooperative and come to speak in a polite way.

2.4. Acoustic Features Caused by the Different Dialogue Partners

Table 6 and Table 7 show the statistics of prosodic features without utterances of correction, reentry and negation. At the beginning, we compared the features between dialogue situations or dialogue partners in each experiment.

According to the result of the T or F statistical tests, the following characteristics were found in each dialogue situation: The mean response time (Start pause) after the change of dialogue turns in the case of human operator is shorter than the other dialogue partners; RMS maximum in the case of WOZ system with a driving task is larger than that of the human operator in *EXP2*; RMS mean in the case of Pseudo or WOZ system is larger than that of the human operator in *EXP2*;

Table 6: *Acoustic Features of EXP1*

Dialogue Situation	O	O+DT	P	P+DT	W	W+DT
Start pause(sec)	0.41	0.31	0.66	0.78	0.63	0.65
Duration(1)(sec)	1.90	1.87	1.75	1.77	1.86	1.92
Duration(2)(sec)	1.37	1.43	1.28	1.37	1.38	1.41
Speaking rate(1)	7.53	7.52	7.74	7.63	7.64	7.22
Speaking rete(2)	10.50	9.65	10.69	9.85	10.26	9.64
RMS mean	606	830	729	919	680	909
RMS max	2676	3548	2963	3544	2701	3475
Pitch mean	71.0	78.4	71.9	80.2	72.7	77.9
Pitch min	57.2	59.2	58.6	60.5	59.0	58.0
Pitch max	245.1	281.3	277.0	251.6	301.1	260.0
Pitch S.D.	15.3	17.5	15.7	17.7	17.5	17.4

Table 7: *Acoustic Features of EXP2*

Dialogue Situation	O	O+DT	P	P+DT	W	W+DT
Start pause(sec)	0.42	0.49	0.75	0.68	0.69	0.63
Duration(1)(sec)	2.08	1.96	1.74	1.78	1.51	1.75
Duration(2)(sec)	1.43	1.47	1.26	1.32	1.16	1.33
Speaking rate(1)	7.29	7.29	7.59	7.53	7.92	7.49
Speaking rete(2)	10.31	9.67	10.24	9.87	10.37	9.66
RMS mean	721	904	1032	1146	1023	1281
RMS max	3255	3941	3880	4590	3902	5397
Pitch mean	70.6	74.7	76.9	79.7	77.0	80.0
Pitch min	55.7	57.6	57.1	58.6	59.5	57.3
Pitch max	299.7	256.4	266.3	231.9	268.8	256.2
Pitch S.D.	20.1	20.3	19.1	20.1	19.1	20.5

From the results, Start pause is affected by the voice quality or the speech style where the expressions are restricted, but the difference whether the dialogue partner is a machine or a human is not observed. We consider that the reason is that it is difficult for users to predict the end of the system utterance without the prosodic information and they are confused about when to start an utterance, or since they feel that it is clearer who will speak, they think that it is not necessary to insist their turn of utterance. Thus they wait for the end of system utterance and begin to speak, so Start pause is longer than that of the human operator. As for start pause of the dialogue partners, mean of this pause in the case of natural voice partner is about 1.3 seconds and in the case of synthetic voice partner, it is 2 seconds. There is a little difference between Start pauses of these partners, but the pauses are found in everyday dialogue and do not probably affect the result. It is found that if sometimes the dialogue partner's speech recognition fails, one tends to speak to Pseudo or WOZ system more loudly than a human operator. And it is, not significant, found that RMS mean and maximum of WOZ-system are the largest in cases of all the dialogue partners except RMS maximum of WOZ/absence, and those of operator are the smallest in cases of all the dialogue partners. This tendency might suggests that RMS is affected by both the voice, natural or synthetic, and the sense whether one thinks the partner as a machine. One tries to keep volume from increasing unconsciously if the partner is a human, probably because the loud voice have to do with the feeling such as anger,

Next, we compared the features between the experiments on each conditions and according to the results of T and F statistical tests, following characteristics were found for the *EXP2* in comparison to the *EXP1*: increase of RMS mean and maximum in the case of WOZ-system ($p < 0.01$) and Pseudo-system except Pseudo/absence ($p < 0.1$); increase of Pitch S.D. in the

case of Operator/absence ($p < 0.01$), Pseudo/absence ($p < 0.05$) and WOZ/existence ($p < 0.1$);

To sum up the results, one tends to speak louder voice to those who sometimes fail speech recognition and this tendency is affected by the voice quality such as natural or synthetic. This may be why synthetic voice makes an impression that the partner is a machine and makes one speak as if talking to a machine. And lower speech recognition rate increases Pitch S.D., which suggests that one tries to intonate own utterances to be understood easily.

2.5. Linguistic Features Caused by a Concurrent Task

As discussed in section 2.3, some linguistic characteristics were found in regard to the difference of dialogue partners. However, with respect to the existence/absence of the concurrent driving task, no significant linguistic differences were observed. One hypothesis is that some cognitive load like car-driving does not affect functions of the human mind like language processing. This hypothesis is found in [7] and the results of study [2] also bear it out.

2.6. Acoustic Features Caused by a Concurrent Task

As for acoustic features, the following characteristics caused by a concurrent task were found for the EXP2 in comparison to the *EXP1* according to the result of the T or F statistical tests: The concurrent task decreases speaking rate except the pause (Speaking rate(2)) in both experiments; the task increases RMS mean in all the dialogue partner cases ($p < 0.1$) in *EXP1*; the task increases Pitch mean in the case of a human or a Pseudo-system ($p < 0.1$) in *EXP1*; the task increases RMS maximum in the case of a human or a WOZ-system ($p < 0.05$) in *EXP2*;

The concurrent car-driving task increases mainly "RMS" and "Pitch mean" in the case of all the dialogue partners. These facts suggest that the psychological load such as a car-driving task affects phonatory organ. The reason may be that the psychological load makes the users to be strained. While Speaking rate(2) is also affected by the existence/absence of the concurrent task, Speaking rate(1) is not much affected. This fact suggests that with the psychological load, one should pause longer and speak more slowly.

2.7. Linguistic and Acoustic Features of Correction Utterances

When we compared correction utterances with other utterances, few significant differences were observed in T and F statistical tests results. However (the table of results is omitted due to space limitation) the following tendencies are found in the all dialogue partners: decrease of keywords per utterance; increase of RMS mean and max; increase of RMS S.D.;

The decrease of keywords per utterance suggests that he/she tends to talk in fewer keywords when one discovers that the dialogue partner sometimes fails in speech recognition. The increase of RMS indicates that one tends to talk in loud voice, and that of Pitch S.D. indicates that one tends to emphasize intonation in such situation. If the partner actually makes a mistake, these tendencies become stronger.

2.8. Linguistic Features Caused by the Landmark Setting task

In this experiment, the users tend to enter whole place-name or only the goal at first rather than gradually enter the detailed place-name. In other study of the same kind[2], the users tended not to enter whole place-name in the case of the machine partner. It may be caused by memorization of all the goals. However, since the goals are known normally this tendency probably appears in the real environment.

3. Conclusions

In this study, we investigated the influence of different dialogue situations, partners and condition patterns of speech recognition rate on the linguistic and acoustic features. The dialogue partners included a human, a machine, and a human who is allowed to do only what the machine can do. We analyzed the overall characteristics of the linguistic and acoustic features in terms of the intra-utterance and the whole utterance statistics. As a result, some linguistic features were affected by whether the responses were the natural voice or the synthetic voice, and some were affected by the speech recognition rate. As for acoustic features, some features were also affected by the difference of voice and the speech recognition rate, on the other hand, features are affected by a car-driving task or whether the dialogue partner is a human or a machine.

The features of utterances are different between in the cases of a human and a machine. In other words, in the case of a machine, it cannot talk naturally like in human-human dialogue. It is probably caused by the fact, that its voice quality and rhythm of dialogue are worse than those of a human's. Thus, in our future work, we are going to investigate which factor is significant to realize a natural dialogue resembling human-human conversation, and make a system which focuses on dialogue rhythm.

4. References

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